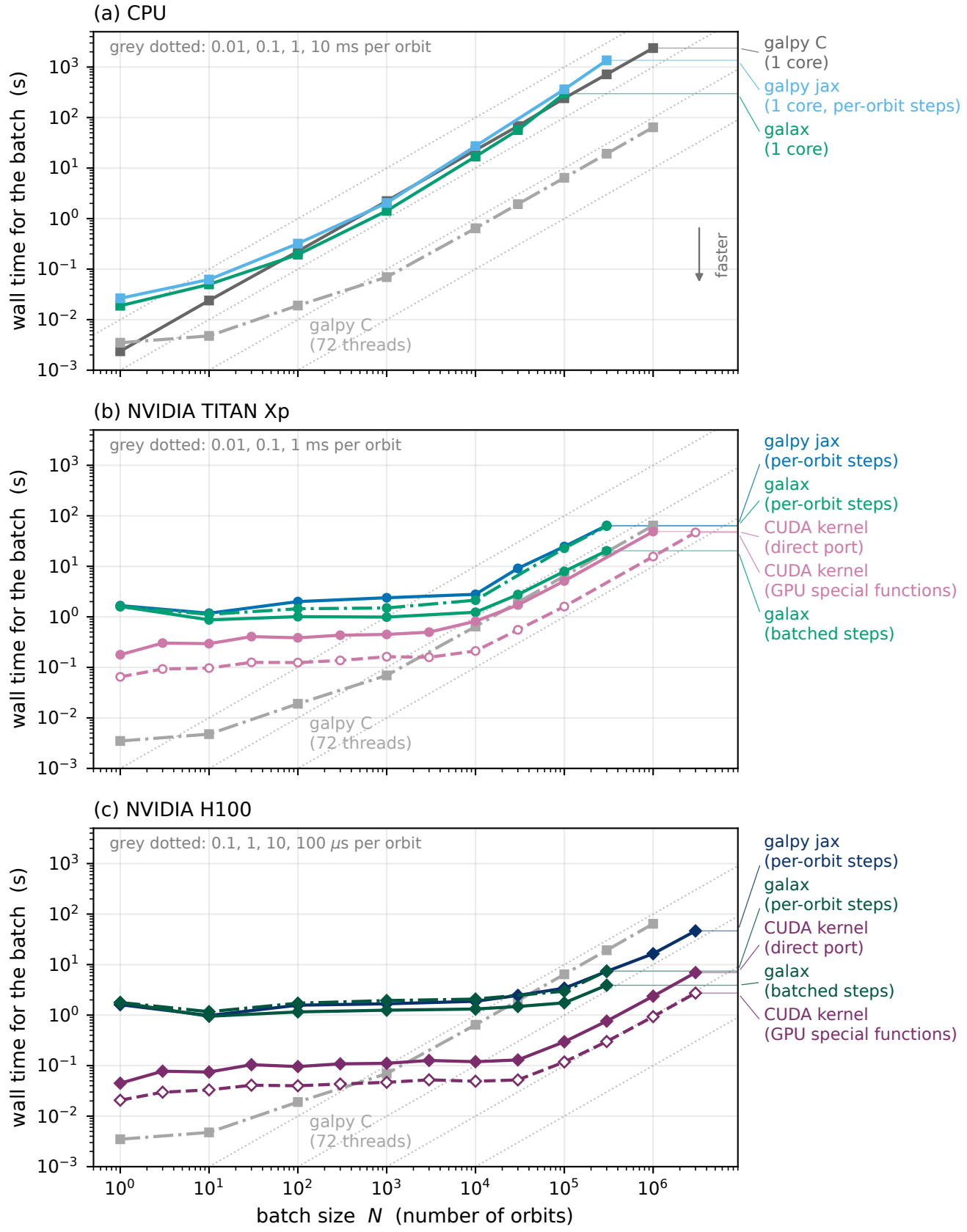
(a) CPU
grey dotted: 0.01, 0.1, 1, 10 ms per orbit
galpy C
(1 core)
galpy jax
(1 core, per-orbit steps)
galax
(1 core)
faster
galpy C
(72 threads)
wall time for the batch (s)
(b) NVIDIA TITAN Xp
grey dotted: 0.01, 0.1, 1 ms per orbit
galpy jax
(per-orbit steps)
galax
(per-orbit steps)
CUDA kernel
(direct port)
CUDA kernel
(GPU special functions)
galax
(batched steps)
galpy C
(72 threads)
wall time for the batch (s)
(c) NVIDIA H100
grey dotted: 0.1, 1, 10, 100 μs per orbit
galpy jax
(per-orbit steps)
galax
(per-orbit steps)
CUDA kernel
(direct port)
galax
(batched steps)
CUDA kernel
(GPU special functions)
galpy C
(72 threads)
wall time for the batch (s)
batch size N (number of orbits)